

Energy Price Inflation & UK Household Impact - ERD Explanation

Updated in Week 4 after loading the real-world data. Text changed because of the new `PAYMENT_METHOD` entity is marked "(updated W4)" and new content is in section 12. Highlighted table rows are new in Week 4.

1 Define the scope of your societal problem

The goal of our database is to investigate how rising energy prices, inflation, and central-bank policies relate to the financial pressures faced by UK households. The constructed database will allow us to compare household energy spending and recorded borrowing costs across regions and demographic groups. Some of the concrete questions we will attempt to answer are as follows: How do household energy bills change over time and compare among regions? How does energy consumption and spending differ between vulnerable households?

The database supports comparisons, but cannot establish cause and effect by itself or calculate exact energy spending because it stores income brackets rather than exact incomes of households.

2 List how the data interacts in short sentences

- A region can have many macroeconomic records, but each record belongs to one region.
- A region can contain many households, but each household belongs to one region.
- A household can receive many energy bills, but each bill belongs to one household.
- An energy supplier can issue many bills, but each bill references one supplier.
- **A payment method can apply to many energy bills, but each bill has at most one payment method (updated W4).**
- A household can have several demographic profiles, and a profile can describe several households.
- A commodity shock can be linked to several macroeconomic records, and a macroeconomic record can be linked to several shocks.
- A central-bank policy can have recorded impacts on several households, and a household can have recorded impacts from several policies.

3 Identify the entities

The nine core entities are `REGION`, `HOUSEHOLD`, `DEMOGRAPHIC_PROFILE`, `ENERGY_BILL`, `ENERGY_SUPPLIER`, **`PAYMENT_METHOD`**, `MACROECONOMIC_METRIC`, `COMMODITY_SHOCK`, and `CENTRAL_BANK_POLICY` (updated W4: `PAYMENT_METHOD` added). Three bridge entities are `HOUSEHOLD_DEMOGRAPHIC`, `SHOCK_IMPACT_LOG`, and `HOUSEHOLD_POLICY_IMPACT`.

4 Identify the relationships and their cardinalities

Entities	Relationship	Cardinality
Region to Household	Contains	One-to-many
Region to Macroeconomic metric	Has measurements	One-to-many
Household to Energy bill	Receives	One-to-many
Energy supplier to Energy bill	Decides	One-to-many
Payment method to Energy bill (W4)	Is paid by (optional)	One-to-many
Household and Demographic profile	Is classified by	Many-to-many
Commodity shock to Macroeconomic metric	Has an impact on	Many-to-many
Central-bank policy and Household	Has an impact on	Many-to-many

Each child record references exactly one parent through its required FK (the payment method FK in `ENERGY_BILL` is optional, because the mock bills have no recorded payment method). A parent can

exist without a related child. For example, a household can be recorded before any energy bills are added.

5 Bridge tables

- HOUSEHOLD_DEMOGRAPHIC connects households to demographic profiles and records the assignment date.
- SHOCK_IMPACT_LOG connects commodity shocks to macroeconomic records and stores the impact type and value.
- HOUSEHOLD_POLICY_IMPACT connects policies to households and stores borrowing costs and mortgage rates.

Each many-to-many relationship becomes two one-to-many relationships. Together with the other one-to-many relationships this produces a total of 11 relationships shown in the ERD (updated W4: 10 before PAYMENT_METHOD was added).

6 Decide which attributes to include per entity

We need to account for circumstances of households, energy consumptions and spendings, economic conditions and recorded impacts. Billing periods and recording dates allow comparisons across time to draw comparisons.

Entity	Attributes
REGION	region_id, region_name (unique, W4), country, total_population
HOUSEHOLD	household_id, region_id, postcode, dwelling_type, occupant_count, housing_tenure
DEMOGRAPHIC_PROFILE	demographic_id, income_bracket, age_group, vulnerability_flag
HOUSEHOLD_DEMOGRAPHIC	household_id, demographic_id, assigned_date
ENERGY_SUPPLIER	supplier_id, supplier_name, ofgem_license_code, support_scheme_enrolled
PAYMENT_METHOD (W4)	payment_method_id, payment_method_name (unique)
ENERGY_BILL	bill_id, household_id, supplier_id, payment_method_id (W4, nullable), billing_period_start, billing_period_end, kwh_consumed, price_cap_rate (GBP per kWh, DECIMAL(10,6) since W4), total_amount_gbp
MACROECONOMIC_METRIC	metric_id, region_id, recording_date, headline_cpi, inflation_rate, energy_cpi_weight (region_id + recording_date unique, W4)
COMMODITY_SHOCK	shock_id, event_name, commodity_type, shock_start_date, shock_end_date, wholesale_price_index
SHOCK_IMPACT_LOG	shock_id, metric_id, impact_type, impact_value
CENTRAL_BANK_POLICY	policy_id, effective_date, base_interest_rate, mortgage_stress_index
HOUSEHOLD_POLICY_IMPACT	policy_id, household_id, borrowing_cost_gbp, mortgage_rate

7 Primary keys and Foreign keys

Every table has a unique primary key. Each composite bridge key prevents duplicate records. ofgem_license_code is an alternate key and must be unique. Since Week 4, region_name and payment_method_name are alternate keys, and (region_id, recording_date) is an alternate key of MACROECONOMIC_METRIC.

8 ERD in Crow foot notation

Primary and foreign keys

Entity	Primary key	Foreign keys
REGION	region_id	None
HOUSEHOLD	household_id	region_id to REGION
DEMOGRAPHIC_PROFILE	demographic_id	None
HOUSEHOLD_DEMOGRAPHIC	household_id, demographic_id	household_id to HOUSEHOLD; demographic_id to DEMOGRAPHIC_PROFILE
ENERGY_SUPPLIER	supplier_id	None

Entity	Primary key	Foreign keys
PAYMENT_METHOD (W4)	payment_method_id	None
ENERGY_BILL	bill_id	household_id to HOUSEHOLD; supplier_id to ENERGY_SUPPLIER; payment_method_id to PAYMENT_METHOD (W4)
MACROECONOMIC_METRIC	metric_id	region_id to REGION
COMMODITY_SHOCK	shock_id	None
SHOCK_IMPACT_LOG	shock_id, metric_id	shock_id to COMMODITY_SHOCK; metric_id to MACROECONOMIC_METRIC
CENTRAL_BANK_POLICY	policy_id	None
HOUSEHOLD_POLICY_IMPACT	policy_id, household_id	policy_id to CENTRAL_BANK_POLICY; household_id to HOUSEHOLD

The ERD diagram itself (Diagram_ERD.pdf) must be redrawn with the PAYMENT_METHOD entity and the new ENERGY_BILL foreign key.

9 Supertypes and subtypes

Supertypes were decided to not be necessary as the entities describe distinct objects and there is no clear shared parent that would make the design easier.

10 Normalize the ERD

1NF form requires atomic values

Each field should contain one value. Multiple demographic memberships must be stored as separate records.

Before

household_id PK	postcode	demographic_ids (violation)
101	NE1 1AA	201, 202
102	NE1 2BB	201

Violation: household 101 has two demographic IDs in one cell. Household 101 cannot act as a single demographic foreign key, and queries would need to separate the list to locate each membership.

After

household_id PK part	demographic_id PK part	postcode
101	201	NE1 1AA
101	202	NE1 1AA
102	201	NE1 2BB

The fields are now atomic. The pair of household_id and demographic_id identifies each membership. However, the household postcode is repeated, creating the partial dependency.

2NF removes partial dependencies

After 1NF, non-key attributes must depend on the complete key. With a composite key of household_id, demographic_id, household or profile facts that depend on only one ID belong in their parent table.

Before

household_id PK	demographic_id PK	postcode (violation)	income_bracket (violation)	assigned_date
101	201	NE1 1AA	Low	2026-01-01
101	202	NE1 1AA	Low	2026-01-01
102	201	NE1 2BB	Low	2026-02-01

Violations: postcode depends only on household_id and income_bracket depends only on demographic_id.

After HOUSEHOLD

household_id PK	postcode
101	NE1 1AA
102	NE1 2BB

After DEMOGRAPHIC_PROFILE

demographic_id PK	income_bracket	age_group	vulnerability_flag
201	Low	18-64	True
202	Low	65+	True

After HOUSEHOLD_DEMOGRAPHIC

household_id PK FK	demographic_id PK FK	assigned_date
101	201	2026-01-01
101	202	2026-01-01
102	201	2026-02-01

Household and profile information is now stored independently. assigned_date remains in the bridge entity because it describes a particular household-profile.

3NF removes transitive dependencies

After 2NF, transitive dependencies must be removed. Supplier details should depend on the supplier key and be maintained in the supplier table.

Before supplier info repeated on bills

bill_id PK	household_id	supplier_id	supplier_name (violation)
401	101	301	North Energy
402	101	301	North Energy
403	102	301	North Energy

Violation: bill_id determines supplier_id, which determines supplier_name. Updating a supplier's name would require changing multiple bill records.

After ENERGY_SUPPLIER

supplier_id PK	supplier_name
301	North Energy

After ENERGY_BILL

bill_id PK	household_id FK	supplier_id FK
401	101	301
402	101	301
403	102	301

Supplier information is stored once, while bills retain the foreign key. This avoids inconsistent supplier names across bills and allows a supplier record to exist even before it has any bills. The same principle supports separating regional details into REGION, shared policy details into CENTRAL_BANK_POLICY and (since Week 4) payment method names into PAYMENT_METHOD.

11 ERD still meets the scope requirements

The ERD supports the comparisons through its household, demographic, billing, regional, shock, and policy records. All entities remain connected, including the new PAYMENT_METHOD entity.

12 Normalization check with real-world data (Week 4, new)

Result: with the real data the database was **not** fully in 3NF at first. Loading the two real datasets exposed one 1NF violation, two 2NF violations and two 3NF violations. Four of them came from the flat shape of the source files and from how the first version of seed_data_real.sql fitted the data into

the tables. One (the derived bill total) is a documented exception. After the fixes in 12.4 every table is in 3NF.

12.1 Data that was loaded

Source file	Rows	Loaded into
UK_ONS_inflation_2000_2025.csv (monthly UK CPI, ONS)	312	MACROECONOMIC_METRIC (+ REGION)
UK_energy_prices_2000_2025.csv (yearly domestic electricity bills, DESNZ QEP 2.2.3, 3 payment methods)	78	ENERGY_BILL (+ PAYMENT_METHOD, ENERGY_SUPPLIER, HOUSEHOLD, REGION)

Both files only exist at UK level, so they use one region (United Kingdom). DEMOGRAPHIC_PROFILE, HOUSEHOLD_DEMOGRAPHIC, COMMODITY_SHOCK, SHOCK_IMPACT_LOG, CENTRAL_BANK_POLICY and HOUSEHOLD_POLICY_IMPACT are not covered by these files and keep their mock data.

12.2 Normal form violations that were present

#	Where	Normal form	Violation	Resolution
V1	Both CSV files (flat)	2NF	region_name → country. Country depends on only part of the key (region_name, recording_date in the CPI file; region_name, recording_year, commodity_type, payment_method in the price file), so it is repeated in every row.	Stored once in REGION. Both loads only use region_id.
V2	Price CSV (flat)	2NF	(commodity_type, recording_year) → standard_consumption_kwh (3300 kWh for 2000-2012, 3800 for 2013-2019, 3600 for 2020-2021, 3400 for 2022-2025). Also recording_year → source_sheet, source_row. These depend on part of the key only.	kwh_consumed is a per-bill fact in ENERGY_BILL (key bill_id), so no partial dependency remains. source_sheet and source_row are provenance metadata and are not loaded.
V3	First seed_data_real.sql, HOUSEHOLD	1NF	dwelling_type = “National average, credit customers” packs two facts into one value (aggregation level + payment method), and postcode = “UK-CREDIT” is a code, not a postcode. The payment method could only be found by parsing text.	New table PAYMENT_METHOD. ENERGY_BILL.payment_method_id holds the payment method as one atomic value.
V4	First seed_data_real.sql, HOUSEHOLD	3NF	The same payment-method fact was stored twice in a household row, so non-key postcode determined non-key dwelling_type (UK-CREDIT → “... credit customers”). Three “households” existed only to carry a payment method.	One representative household (UK-AVG). Payment method moved to the bill.
V5	ENERGY_BILL (mock and real data)	3NF (derived attribute)	(kwh_consumed, price_cap_rate) → total_amount_gbp (total ≈ kWh × rate). Non-key attributes determine another non-key attribute.	Kept deliberately, see 12.5.

Data-quality problems found while loading (not normal-form violations, fixed or documented):

- **Unit mismatch in price_cap_rate.** The real CSV has pence per kWh (e.g. 7.77) while the mock data has GBP per kWh (e.g. 0.21). Real values are now divided by 100 and the column was widened from DECIMAL(10,4) to DECIMAL(10,6) to keep the source precision.
- **Values that are not in the sources.** The first load invented total_population = 67,000,000; it is now NULL. support_scheme_enrolled = FALSE and the postcode sentinel UK-AVG are placeholders because those columns are NOT NULL.
- **Wrong source label.** The first load called the data “Ofgem”. The source is DESNZ (QEP table 2.2.3), as the README states.
- **Missing candidate keys.** UNIQUE(region_name), UNIQUE(region_id, recording_date) and a unique key on bills were added so the dependencies above are enforced by the database.
- **energy_cpi_weight** is not published by ONS, so all 312 values are NULL.

12.3 Before and after

V3/V4 - payment method hidden in the household (before)

household_id	region_id	postcode	dwelling_type
31	9	UK-CREDIT	National average, credit customers
32	9	UK-DD	National average, direct debit customers
33	9	UK-PPM	National average, prepayment customers

After PAYMENT_METHOD

payment_method_id PK	payment_method_name
1	Credit
2	Direct debit
3	Prepayment

After HOUSEHOLD (one row): 31 | region_id 9 | UK-AVG | NULL

After ENERGY_BILL (excerpt)

bill_id	household_id	payment_method_id FK	billing_period_start	kwh_consumed	price_cap_rate (GBP/kWh)	total_amount_gbp
46	31	1	2000-01-01	3300.00	0.077700	257.00
72	31	2	2000-01-01	3300.00	0.074200	245.00
98	31	3	2000-01-01	3300.00	0.083100	274.00

Payment method names are stored once. Renaming one is a single update, and a payment method can exist before it has any bills.

V1 - country repeated (before, flat CPI file)

region_name	recording_date	country	headline_cpi
United Kingdom	2000-01-01	United Kingdom	71.9
United Kingdom	2000-02-01	United Kingdom	72.2

After: REGION(9, United Kingdom, United Kingdom, NULL) and MACROECONOMIC_METRIC(region_id 9, 2000-01-01, 71.9, ...). Country is stored once.

12.4 Changes made to the ERD and schema

- New entity PAYMENT_METHOD(payment_method_id PK, payment_method_name UNIQUE NOT NULL) with an optional one-to-many relationship to ENERGY_BILL.
- ENERGY_BILL: new nullable column payment_method_id (FK). price_cap_rate becomes DECIMAL(10,6) in GBP per kWh.
- New UNIQUE constraints on REGION(region_name), MACROECONOMIC_METRIC(region_id, recording_date) and ENERGY_BILL(household_id, supplier_id, payment_method_id, billing_period_start).
- The real data is inserted with seed_data_real.sql (generated from the CSV files) and checked with verify_normalization.sql.

12.5 Documented exceptions and observations

- **total_amount_gbp is a derived attribute (V5).** Strictly, this is a transitive dependency. We keep it because it is the published value in the source and queries 1 and 2 sum it. It is also not an exact function of the other two columns: 66 of 78 real bills match kWh × rate to the penny, 11 differ by less than £0.50, and one (2000, Credit) differs by £0.59 because the source rounds the early years to whole pounds. If a strict 3NF is required, the alternative is to drop the column and compute it in a view.
- **Repeated kwh_consumed.** For each year the three payment methods share the same standard consumption, so the value is repeated three times. This is data redundancy, not a 3NF violation, because the rule “same year ⇒ same kWh” is a property of this source and not of the schema; in the mock data, bills of the same period have different kWh.
- **household_id → postcode → region_id.** In reality a postcode lies in one region. This comes from the Week 2 design and not from the real data (the real household uses a sentinel postcode).

It is kept because postcodes are not validated against a postcode lookup table.

- **Mixed granularity.** The United Kingdom region overlaps the eight mock regions, so regional comparisons should not add it to the others. Query 1 is not affected (real bills cover whole years, outside Q1 2022). Query 2 also returns real national-average bills; add `payment_method_id` IS NULL to keep only the mock households.

12.6 Result

After loading the real data with the updated `schema.sql` and `seed_data_real.sql`, every table satisfies 1NF (atomic values), 2NF (no partial dependencies) and 3NF (no transitive dependencies), except the documented derived total in 12.5. We checked the load by running schema, mock data, real data and queries in MySQL-compatible MariaDB 10.11 and comparing all 78 bills and 312 CPI rows against the CSV files, with no mismatches.